

Multiple Choice (10 marks)

1. The anhydride of $\text{Ba}(\text{OH})_2$ is
 - (a) BaH_2
 - (b) BaOH
 - (c) Ba
 - (d) BaO
2. Which of the following experimental observations were explained by Planck's quantum theory?
 - (a) Blackbody radiation curves
 - (b) Emission spectra of diatomic molecules
 - (c) Electron diffraction patterns
 - (d) Temperature dependence of reaction rates
3. The solid-state structures of the principal allotropes of elemental boron are made up of which of the following structural units?
 - (a) B_{12} icosahedra
 - (b) B_8 cubes
 - (c) B_6 octahedra
 - (d) B_4 tetrahedra
4. Of the following atoms, which has the lowest electron affinity?
 - (a) F
 - (b) Si
 - (c) O
 - (d) Ca
5. A set of hybrid sp^3 orbitals for a carbon atom is given above. Which of the following is NOT true about the orbitals?
 - (a) The orbitals are degenerate.
 - (b) The set of orbitals has a tetrahedral geometry.
 - (c) These orbitals are constructed from a linear combination of atomic orbitals.
 - (d) Each hybrid orbital may hold four electrons.

True / False (10 marks)

Answer True or False

6. True or False: The maximum number of phases that can be at equilibrium in a three-component mixture is five.
7. True or False: The infrared absorption frequency of deuterium chloride (DCl) is shifted from that of hydrogen chloride (HCl) due to differences in reduced mass.
8. True or False: The NMR frequency of ^{31}P in a 20.0 T magnetic field is 345.0 MHz.
9. True or False: The orbital angular momentum quantum number, l , of the electron most easily removed from ground-state aluminum is 2.
10. True or False: Both paramagnetism and ferromagnetism require the presence of unpaired electrons in atoms or ions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the characteristics of Lewis acids and evaluate why SCl_2 is the least likely to exhibit Lewis acid behavior compared to other compounds in its class.
12. Discuss the role of dopants in semiconductors, specifically focusing on how arsenic-doped silicon functions as an n-type semiconductor and its implications for electronic applications.

End of Paper